

EX 05 - MySQL Fragmentação Horizontal

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Topic: MySQL Fragmentação Horizontal

Lab: [EX5 - MySQL Fragmentação Horizontal.pdf](#)

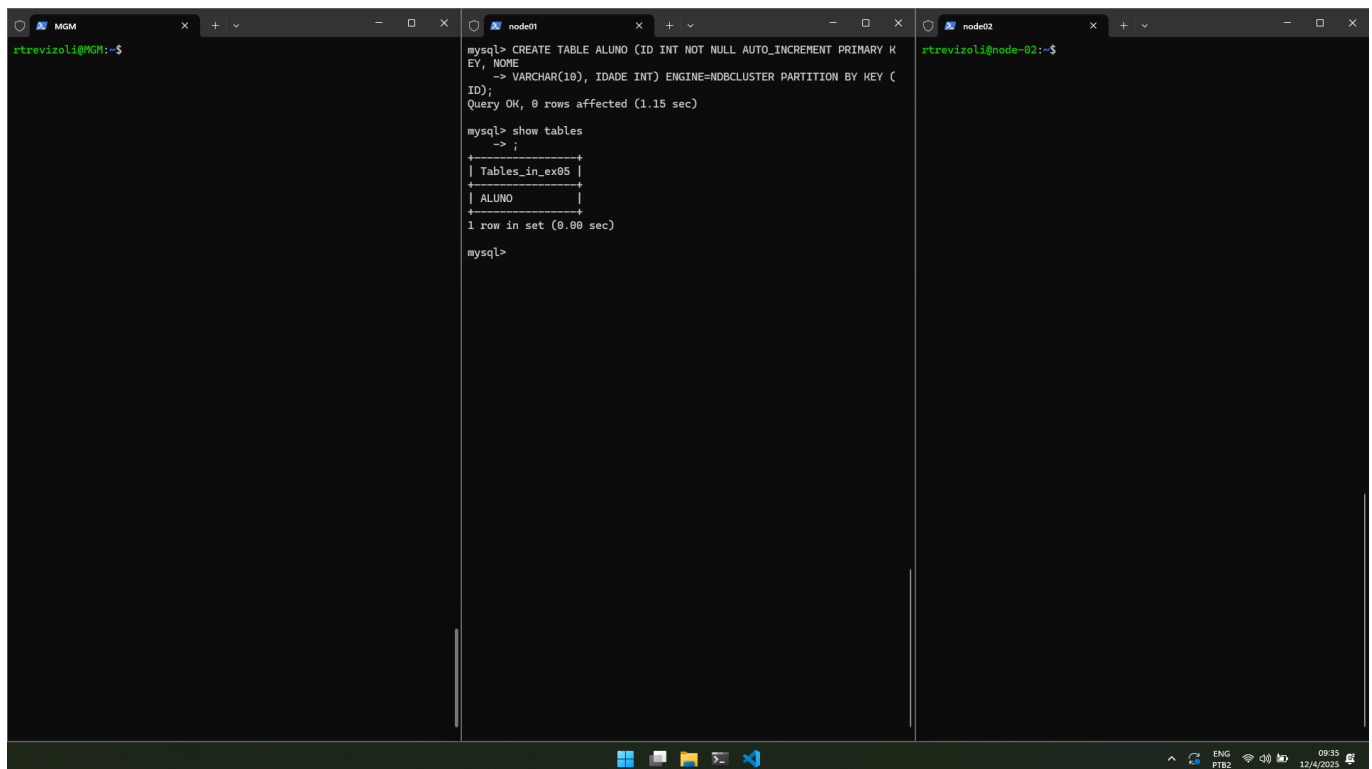
EX5 – MySQL Fragmentação Horizontal

Importante 1: As etapas abaixo descritas estão sendo executadas com todos os três nós ligados e conectados.

Importante 2: Criado schema **ex05** no nó 01 para execução dos passos abaixo.

Fragmentação Horizontal por ID automático

```
CREATE TABLE ALUNO (ID INT NOT NULL AUTO_INCREMENT PRIMARY KEY, NOME  
VARCHAR(10), IDADE INT) ENGINE=NDBCLUSTER PARTITION BY KEY (ID);
```



Img 01 - Create table aluno partioned by ID in node01

1. Insira pelo menos 15 registros

```
INSERT INTO ALUNO (NOME, IDADE) VALUES  
( 'Ana', 20), ( 'Bruno', 22), ( 'Carla', 19), ( 'Daniel', 21), ( 'Eva', 23),
```

```
('Felipe', 20), ('Gabi', 18), ('Hugo', 24), ('Iara', 22), ('Joao', 19),
('Katia', 20), ('Leo', 21), ('Mara', 23), ('Nina', 22), ('Otavio', 24);
```

```
mysql> CREATE TABLE ALUNO (ID INT NOT NULL AUTO_INCREMENT PRIMARY KEY,
--> NOME VARCHAR(10), IDADE INT) ENGINE=InnoDB PARTITION BY KEY (
ID);
Query OK, 0 rows affected (1.15 sec)

mysql> show tables
+-----+
| Tables_in_ex05 |
+-----+
| ALUNO           |
+-----+
1 row in set (0.00 sec)

mysql> INSERT INTO ALUNO (NOME, IDADE) VALUES
--> ('Ana', 20), ('Bruno', 22), ('Carla', 19), ('Daniel', 21), ('Eva',
23),
--> ('Felipe', 20), ('Gabi', 18), ('Hugo', 24), ('Iara', 22), ('Joao',
19),
--> ('Katia', 20), ('Leo', 21), ('Mara', 23), ('Nina', 22), ('Otavio',
24);
Query OK, 15 rows affected (0.10 sec)
Records: 15 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM ALUNO
--> ;
+----+-----+-----+
| ID | NOME  | IDADE |
+----+-----+-----+
| 7  | Gabi  | 18    |
| 3  | Carla | 19    |
| 9  | Iara  | 22    |
| 10 | Joao  | 19    |
| 15 | Otavio| 24    |
| 6  | Felipe| 20    |
| 12 | Leo   | 21    |
| 5  | Eva   | 23    |
| 14 | Nina  | 22    |
| 11 | Katia | 20    |
| 1  | Ana   | 20    |
| 13 | Mara  | 23    |
| 2  | Bruno | 22    |
| 4  | Daniel| 21    |
| 8  | Hugo  | 24    |
+----+-----+-----+
15 rows in set (0.07 sec)

mysql>
```

Img 02 - Insert 15 lines in ex05.ALUNO, node01

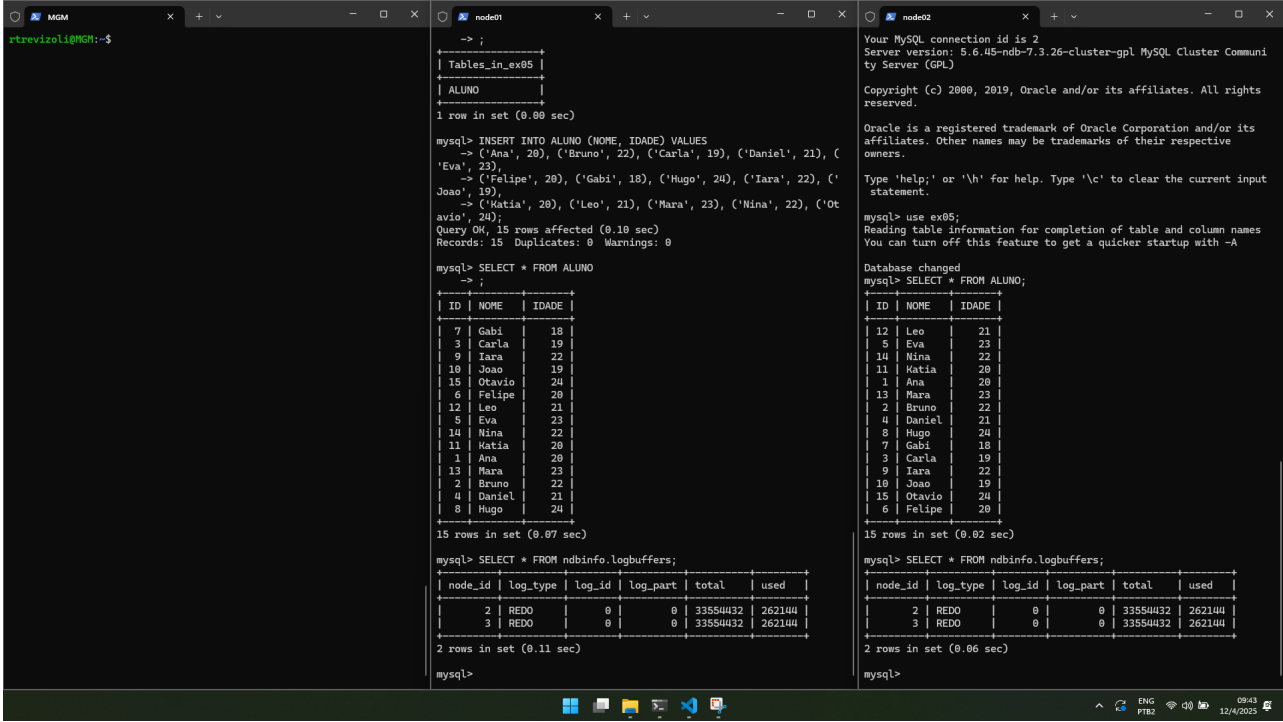
2. Verifique a consistência nos nós

- Alternativa A: Utilizando `ndbinfo.logbuffers` Mostra a consistência de replicação entre os nós.

```
SELECT * FROM ndbinfo.logbuffers;
```

Se as repicações estiverem corretas:

- `Used` não vai crescer indefinidamente;
- Não haverá backlog excessivo;
- Não haverá padrões de "gap"



Img 03 - Check consistency using logbuffers, node01

- Alternativa B: Verificar o status dos nós no **MGM** (Nó de gerenciamento) No nó de gerenciamento:

```
ndb_mgm
```

Então:

```
ALL STATUS
```

Se houver inconsistência em uma réplica, será exibido:

- **not started**;
- **start phase...**;
- **NR** (Not-Recovering);
- **SUSPECT** ou **FAILED**.

The screenshot shows a terminal window with three panes. The left pane shows the MySQL Cluster Management Client (ndb_mgm) interface. The middle pane shows a MySQL session on node01 where data is inserted into the ALUNO table and then queried. The right pane shows a MySQL session on node02 where the same data is queried.

```

ndb_mgm>
ndb_mgm> ALL STATUS
Connected to Management Server at: localhost:1186
Node 2: started (mysql-5.6.45 ndb-7.3.26)
Node 3: started (mysql-5.6.45 ndb-7.3.26)

ndb_mgm>

mysql> INSERT INTO ALUNO (NOME, IDADE) VALUES
-> ('Ana', 20), ('Bruno', 22), ('Carla', 19), ('Daniel', 21), ('Eva', 23),
-> ('Felipe', 20), ('Gabi', 18), ('Hugo', 24), ('Iara', 22), ('Joao', 19),
-> ('Katia', 20), ('Leo', 21), ('Mara', 23), ('Nina', 22), ('Otavio', 24);
Query OK, 15 rows affected (0.18 sec)
Records: 15 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM ALUNO
->
+----+-----+-----+
| ID | NOME | IDADE |
+----+-----+-----+
| 7  | Gabi  | 18    |
| 3  | Carla | 19    |
| 9  | Iara  | 22    |
| 10 | Joao  | 19    |
| 15 | Otavio| 24    |
| 6  | Felipe| 20    |
| 12 | Leo   | 21    |
| 5  | Eva   | 23    |
| 14 | Nina  | 22    |
| 11 | Katia | 20    |
| 1  | Ana   | 20    |
| 13 | Mara  | 23    |
| 2  | Bruno | 22    |
| 4  | Daniel| 21    |
| 8  | Hugo  | 24    |
+----+-----+-----+
15 rows in set (0.07 sec)

mysql> SELECT * FROM ndbinfo.logbuffers;
+----+-----+-----+-----+-----+-----+
| node_id | log_type | log_id | log_part | total | used |
+----+-----+-----+-----+-----+-----+
| 2       | REDO     | 0      | 0        | 33554432 | 262144 |
| 3       | REDO     | 0      | 0        | 33554432 | 262144 |
+----+-----+-----+-----+-----+-----+
2 rows in set (0.11 sec)

mysql>

```

```

Your MySQL connection id is 2
Server version: 5.6.45-ndb-7.3.26-cluster-gpl MySQL Cluster Community Server (GPL)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use ex05;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM ALUNO;
+----+-----+-----+
| ID | NOME | IDADE |
+----+-----+-----+
| 12 | Leo   | 21    |
| 5  | Eva   | 23    |
| 14 | Nina  | 22    |
| 11 | Katia | 20    |
| 1  | Ana   | 20    |
| 13 | Mara  | 23    |
| 2  | Bruno | 22    |
| 4  | Daniel| 21    |
| 8  | Hugo  | 24    |
| 7  | Gabi  | 18    |
| 3  | Carla | 19    |
| 9  | Iara  | 22    |
| 10 | Joao  | 19    |
| 15 | Otavio| 24    |
| 6  | Felipe| 20    |
+----+-----+-----+
15 rows in set (0.02 sec)

mysql> SELECT * FROM ndbinfo.logbuffers;
+----+-----+-----+-----+-----+-----+
| node_id | log_type | log_id | log_part | total | used |
+----+-----+-----+-----+-----+-----+
| 2       | REDO     | 0      | 0        | 33554432 | 262144 |
| 3       | REDO     | 0      | 0        | 33554432 | 262144 |
+----+-----+-----+-----+-----+-----+
2 rows in set (0.06 sec)

mysql>

```

Img 04 - Check consistency using ndb_mgm, mgm

3. Como verificar o que foi armazenado em cada nó?

3.1. O comando abaixo mostrará quantos registros estão alocados fisicamente em cada nó

```
select partition_name, table_rows from information_schema.PARTITIONS
where table_name = 'ALUNO';
```

The screenshot shows a terminal window with three panes. The left pane shows the MySQL Cluster Management Client (ndb_mgm) interface. The middle pane shows a MySQL session on node01 where the query is executed. The right pane shows a MySQL session on node02 where the same query is executed.

```

ndb_mgm>
ndb_mgm> ALL STATUS
Connected to Management Server at: localhost:1186
Node 2: started (mysql-5.6.45 ndb-7.3.26)
Node 3: started (mysql-5.6.45 ndb-7.3.26)

ndb_mgm>

rtrevizoli@node01:~$ select partition_name, table_rows from information_schema.PARTITIONS
-bash: syntax error near unexpected token `table_rows'
rtrevizoli@node01:~$ where table_name = 'ALUNO';^C
rtrevizoli@node01:~$ ^C
rtrevizoli@node01:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 7
Server version: 5.6.45-ndb-7.3.26-cluster-gpl MySQL Cluster Community Server (GPL)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> select partition_name, table_rows from information_schema.PARTITIONS
-> where table_name = 'ALUNO';
+-----+-----+
| partition_name | table_rows |
+-----+-----+
| p0             | 6          |
| p1             | 9          |
+-----+-----+
2 rows in set (0.11 sec)

mysql>

```

```

mysql> SELECT * FROM ALUNO;
+----+-----+-----+
| ID | NOME | IDADE |
+----+-----+-----+
| 12 | Leo   | 21    |
| 5  | Eva   | 23    |
| 14 | Nina  | 22    |
| 11 | Katia | 20    |
| 1  | Ana   | 20    |
| 13 | Mara  | 23    |
| 2  | Bruno | 22    |
| 4  | Daniel| 21    |
| 8  | Hugo  | 24    |
| 7  | Gabi  | 18    |
| 3  | Carla | 19    |
| 9  | Iara  | 22    |
| 10 | Joao  | 19    |
| 15 | Otavio| 24    |
| 6  | Felipe| 20    |
+----+-----+-----+
15 rows in set (0.02 sec)

mysql> select partition_name, table_rows from information_schema.PARTITIONS
-> where table_name = 'ALUNO';
+-----+-----+
| partition_name | table_rows |
+-----+-----+
| p0             | 6          |
| p1             | 9          |
+-----+-----+
2 rows in set (0.25 sec)

mysql>

```

Img 05 - Verify what were the inserts in each node

3.2. O comando abaixo mostrará para cada registro onde ele está armazenado

```
explain partitions select * from ALUNO where id = 1;
```

```

MGM
rtrevizoli@MGM:~$ ndb_mgm
-- NDB Cluster -- Management Client --
ndb_mgm> ALL STATUS
Connected to Management Server at: localhost:1186
Node 2: started (mysql-5.6.45 ndb-7.3.26)
Node 3: started (mysql-5.6.45 ndb-7.3.26)
ndb_mgm>

node01
-bash: syntax error near unexpected token `table_rows'
rtrevizoli@node-01:~$ where table_name = 'ALUNO';^C
rtrevizoli@node-01:~$ ^C
rtrevizoli@node-01:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 7
Server version: 5.6.45-ndb-7.3.26-cluster-gpl MySQL Cluster Community Server (GPL)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> select partition_name, table_rows from information_schema.PARTITIONS
-> where table_name = 'ALUNO';
+-----+-----+
| partition_name | table_rows |
+-----+-----+
| p0             | 6          |
| p1             | 9          |
+-----+-----+
2 rows in set (0.11 sec)

mysql> explain partitions select * from ALUNO where id = 1;
ERROR 1846 (3D000): No database selected
mysql> use ex05;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> explain partitions select * from ALUNO where id = 1;
+----+-----+-----+-----+-----+-----+
| id | select_type | table | partitions | type | possible_keys |
+----+-----+-----+-----+-----+-----+
| 1  | SIMPLE     | ALUNO | p1         | PRIMARY | 4             |
+----+-----+-----+-----+-----+-----+
1 row in set (0.03 sec)

mysql>

node02
mysql> SELECT * FROM ALUNO;
+----+-----+-----+
| ID | NOME  | IDADE |
+----+-----+-----+
| 12 | Leo   | 21    |
| 5  | Eva   | 23    |
| 14 | Nina  | 22    |
| 11 | Katia | 20    |
| 1  | Ana   | 20    |
| 13 | Mara  | 23    |
| 2  | Bruno | 22    |
| 4  | Daniel| 21    |
| 8  | Hugo  | 24    |
| 7  | Gabi  | 18    |
| 3  | Carla | 19    |
| 9  | Iara   | 22    |
| 10 | Joao  | 19    |
| 15 | Otavio| 20    |
| 6  | Felipe| 20    |
+----+-----+-----+
15 rows in set (0.02 sec)

mysql> select partition_name, table_rows from information_schema.PARTITIONS
-> where table_name = 'ALUNO';
+-----+-----+
| partition_name | table_rows |
+-----+-----+
| p0             | 6          |
| p1             | 9          |
+-----+-----+
2 rows in set (0.25 sec)

mysql> use ex05;
Database changed
mysql> explain partitions select * from ALUNO where id = 1;
+----+-----+-----+-----+-----+-----+-----+
| id | select_type | table | partitions | type | possible_keys |
+----+-----+-----+-----+-----+-----+-----+
| 1  | SIMPLE     | ALUNO | p1         | PRIMARY | 4             |
+----+-----+-----+-----+-----+-----+-----+
1 row in set (0.04 sec)

mysql>
  
```

Img 06 - Explain partitions on table ALUNO

4. Faça as mesmas verificações para uma tabela replicada por completo.

Tabela replicada por completo:

```

CREATE TABLE FRT_ALUNO (
  ID INT NOT NULL AUTO_INCREMENT PRIMARY KEY,
  NOME VARCHAR(10),
  IDADE INT
) ENGINE=NDBCLUSTER
COMMENT='NDB_TABLE=FULLY_REPLICATED';
  
```

```

PS C:\Users\rtrevizoli\Projects\Fatec\tbi005> ssh rtrevizoli@172.20.10.7
rtrevizoli@172.20.10.7's password:
Welcome to Ubuntu 16.04.6 LTS (GNU/Linux 4.4.0-210-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

77 packages can be updated.
1 update is a security update.

Last login: Thu Dec 4 11:46:24 2025
rtrevizoli@MGM:~$ ndb_mgm
-- NDB Cluster -- Management Client --
ndb_mgm> ALL STATUS
Connected to Management Server at: localhost:1186
Node 2: started (mysql-5.6.45 ndb-7.3.26)
Node 3: started (mysql-5.6.45 ndb-7.3.26)

ndb_mgm>

Welcome to Ubuntu 16.04.6 LTS (GNU/Linux 4.4.0-142-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

190 packages can be updated.
134 updates are security updates.

Last login: Thu Dec 4 12:53:23 2025 from 172.20.10.2
rtrevizoli@node-01:~$ sudo mysql
[sudo] password for rtrevizoli:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 2
Server version: 5.6.45-ndb-7.3.26-cluster-gpl MySQL Cluster Community Server (GPL)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use ex05;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> CREATE TABLE FRT_ALUNO (
  ->   ID INT NOT NULL AUTO_INCREMENT PRIMARY KEY,
  ->   NOME VARCHAR(10),
  ->   IDADE INT
  -> ) ENGINE=NDBCLUSTER
  -> COMMENT='NDB_TABLE=FULLY_REPLICATED';
Query OK, 0 rows affected, 1 warning (1.12 sec)

mysql> show tables;
+-----+
| Tables_in_ex05 |
+-----+
| ALUNO           |
| FRT_ALUNO       |
+-----+
2 rows in set (0.01 sec)

mysql>

Welcome to Ubuntu 16.04.7 LTS (GNU/Linux 4.4.0-210-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Dec 4 09:27:27 2025 from 172.20.10.2
rtrevizoli@node-02:~$ sudo mysql
[sudo] password for rtrevizoli:
rtrevizoli@node-02:~$ sudo mysql
[sudo] password for rtrevizoli:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 2
Server version: 5.6.45-ndb-7.3.26-cluster-gpl MySQL Cluster Community Server (GPL)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show tables from ex05;
+-----+
| Tables_in_ex05 |
+-----+
| ALUNO           |
| FRT_ALUNO       |
+-----+
2 rows in set (0.04 sec)

mysql>

```

Img 07 - Create table FRT_ALUNO full replicated in node 01

- 1. Insira pelo menos 15 registros

```

INSERT INTO FRT_ALUNO (NOME, IDADE) VALUES
('Ana', 20), ('Bruno', 22), ('Carla', 19), ('Daniel', 21), ('Eva', 23),
('Felipe', 20), ('Gabi', 18), ('Hugo', 24), ('Iara', 22), ('Joao', 19),
('Katia', 20), ('Leo', 21), ('Mara', 23), ('Nina', 22), ('Otavio', 24);

```

```

PS C:\Users\rtrevizoli\Projects\Fatec\tbi005> ssh rtrevizoli@172.20.10.7
rtrevizoli@172.20.10.7's password:
Welcome to Ubuntu 16.04.6 LTS (GNU/Linux 4.4.0-210-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

77 packages can be updated.
1 update is a security update.

Last login: Thu Dec 4 11:46:24 2025
rtrevizoli@MGM:~$ ndb_mgm
-- NDB Cluster -- Management Client --
ndb_mgm> ALL STATUS
Connected to Management Server at: localhost:1186
Node 2: started (mysql-5.6.45 ndb-7.3.26)
Node 3: started (mysql-5.6.45 ndb-7.3.26)

ndb_mgm>

mysql> SELECT * FROM FRT_ALUNO;
Empty set (0.00 sec)

mysql> INSERT INTO FRT_ALUNO (NOME, IDADE) VALUES
  -> ('Ana', 20), ('Bruno', 22), ('Carla', 19), ('Daniel', 21), ('Eva', 23),
  -> ('Felipe', 20), ('Gabi', 18), ('Hugo', 24), ('Iara', 22), ('Joao', 19),
  -> ('Katia', 20), ('Leo', 21), ('Mara', 23), ('Nina', 22), ('Otavio', 24);
Query OK, 15 rows affected (0.05 sec)
Records: 15 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM FRT_ALUNO;
+----+-----+-----+
| ID | NOME  | IDADE |
+----+-----+-----+
| 7  | Gabi  | 18    |
| 3  | Carla | 19    |
| 9  | Iara  | 22    |
| 10 | Joao  | 19    |
| 15 | Otavio | 24    |
| 6  | Felipe | 20    |
| 12 | Leo   | 21    |
| 5  | Eva   | 23    |
| 14 | Nina  | 22    |
| 11 | Katia | 20    |
| 1  | Ana   | 20    |
| 13 | Mara  | 23    |
| 2  | Bruno | 22    |
| 4  | Daniel | 21    |
| 8  | Hugo  | 24    |
+----+-----+-----+
15 rows in set (0.02 sec)

mysql>

mysql> SELECT * FROM FRT_ALUNO;
Empty set (0.04 sec)

mysql> SELECT * FROM FRT_ALUNO
  -> ;
+----+-----+-----+
| ID | NOME  | IDADE |
+----+-----+-----+
| 7  | Gabi  | 18    |
| 3  | Carla | 19    |
| 9  | Iara  | 22    |
| 10 | Joao  | 19    |
| 15 | Otavio | 24    |
| 6  | Felipe | 20    |
| 12 | Leo   | 21    |
| 5  | Eva   | 23    |
| 14 | Nina  | 22    |
| 11 | Katia | 20    |
| 1  | Ana   | 20    |
| 13 | Mara  | 23    |
| 2  | Bruno | 22    |
| 4  | Daniel | 21    |
| 8  | Hugo  | 24    |
+----+-----+-----+
15 rows in set (0.01 sec)

mysql>

```

Img 08 - Insert 15 lines in ex05.FRT_ALUNO, node01

- 2. Verifique a consistência nos nós

- Alternativa A: Utilizando `ndbinfo.logbuffers`
- Alternativa B: Verificar o status dos nós no **MGM** (Nó de gerenciamento)

The screenshot shows three terminal windows. The left window is the Management Client (MGM) showing the status of the NDB Cluster. The middle window is node01, and the right window is node02. Both node windows show the execution of a SQL query to check the consistency of the FRT_ALUNO table and the logbuffers.

node01 Results:

```
mysql> SELECT * FROM FRT_ALUNO;
```

ID	NOME	IDADE
7	Gabi	18
3	Carla	19
9	Iara	22
10	Joao	19
15	Otavio	24
6	Felipe	20
12	Leo	21
5	Eva	23
14	Nina	22
11	Katia	20
1	Ana	20
13	Mara	23
2	Bruno	22
4	Daniel	21
8	Hugo	24

```
mysql> SELECT * FROM ndbinfo.logbuffers;
```

node_id	log_type	log_id	log_part	total	used
2	REDO	0	0	33554432	262144
3	REDO	0	0	33554432	262144

node02 Results:

```
mysql> SELECT * FROM FRT_ALUNO;
```

ID	NOME	IDADE
12	Leo	21
5	Eva	23
14	Nina	22
11	Katia	20
1	Ana	20
13	Mara	23
2	Bruno	22
4	Daniel	21
8	Hugo	24
7	Gabi	18
3	Carla	19
9	Iara	22
10	Joao	19
15	Otavio	24
6	Felipe	20

```
mysql> SELECT * FROM ndbinfo.logbuffers;
```

node_id	log_type	log_id	log_part	total	used
2	REDO	0	0	33554432	262144
3	REDO	0	0	33554432	262144

Img 09 - Check node consistency using A and B approach

- 3. Como verificar o que foi armazenado em cada nó?
 - 3.1. O comando abaixo mostrará quantos registros estão alocados fisicamente em cada nó

The screenshot shows three terminal windows. The left window is the Management Client (MGM) showing the status of the NDB Cluster. The middle window is node01, and the right window is node02. Both node windows show the execution of a SQL query to check the consistency of the FRT_ALUNO table and the logbuffers, and then a query to select the partition name and table rows from the information_schema.PARTITIONS table.

node01 Results:

```
mysql> SELECT * FROM FRT_ALUNO;
```

ID	NOME	IDADE
7	Gabi	18
3	Carla	19
9	Iara	22
10	Joao	19
15	Otavio	24
6	Felipe	20
12	Leo	21
5	Eva	23
14	Nina	22
11	Katia	20
1	Ana	20
13	Mara	23
2	Bruno	22
4	Daniel	21
8	Hugo	24

```
mysql> SELECT * FROM ndbinfo.logbuffers;
```

node_id	log_type	log_id	log_part	total	used
2	REDO	0	0	33554432	262144
3	REDO	0	0	33554432	262144

```
mysql> select partition_name, table_rows from information_schema.PARTITIONS
-> where table_name = 'FRT_ALUNO';
```

partition_name	table_rows
p0	6
p1	9

node02 Results:

```
mysql> SELECT * FROM FRT_ALUNO;
```

ID	NOME	IDADE
12	Leo	21
5	Eva	23
14	Nina	22
11	Katia	20
1	Ana	20
13	Mara	23
2	Bruno	22
4	Daniel	21
8	Hugo	24
7	Gabi	18
3	Carla	19
9	Iara	22
10	Joao	19
15	Otavio	24
6	Felipe	20

```
mysql> SELECT * FROM ndbinfo.logbuffers;
```

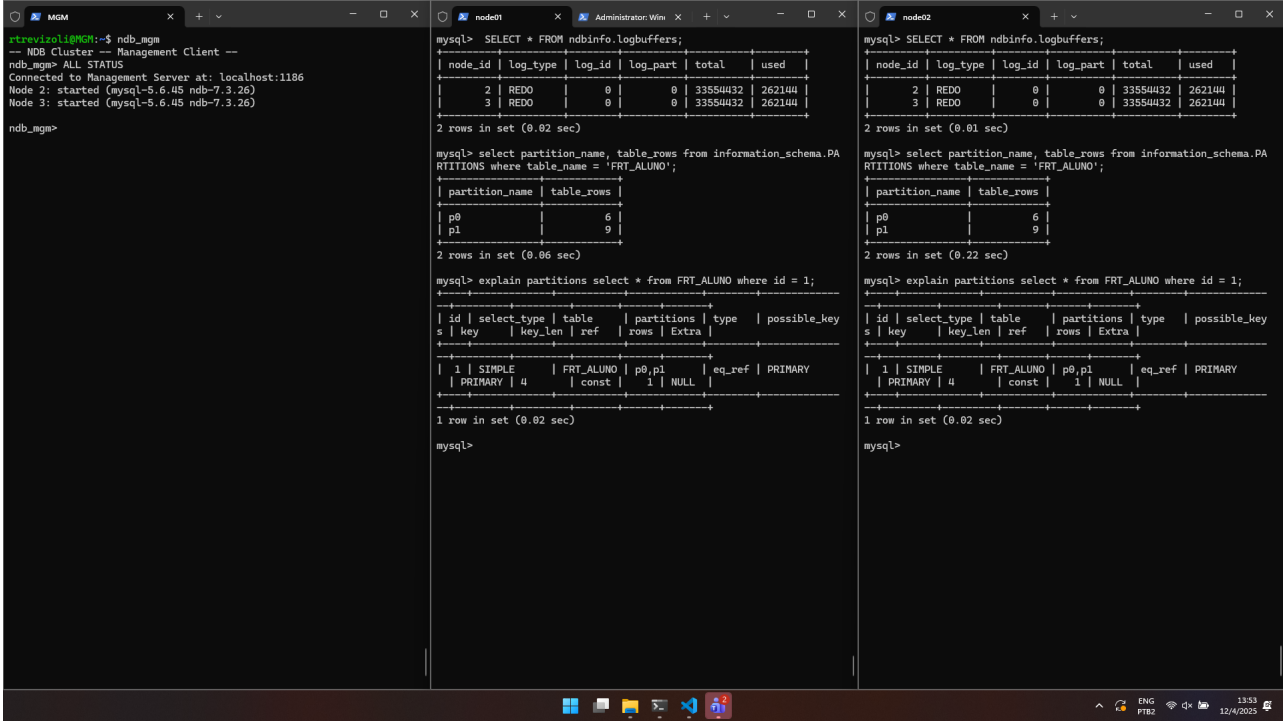
node_id	log_type	log_id	log_part	total	used
2	REDO	0	0	33554432	262144
3	REDO	0	0	33554432	262144

```
mysql> select partition_name, table_rows from information_schema.PARTITIONS
-> where table_name = 'FRT_ALUNO';
```

partition_name	table_rows
p0	6
p1	9

Img 10 - Verify what were the inserts in each node

- 3.2. O comando abaixo mostrará para cada registro onde ele está armazenado



Img 11 - Explain partitions on table FRT_ALUNO